

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed **event-driven backend infrastructure** for a distributed access control platform, processing over 1 million device events daily to enable real-time coordination and reliability.
- Managed **large-scale Linux device fleet** operations and updates, coordinating rollouts to over 5,000 deployed controllers across unreliable network conditions, achieving a 99.8% successful update rate.
- Designed and implemented **device identity and authentication systems** using secure communication protocols, ensuring robust security for cloud-connected access control devices.
- Engineered **backend APIs and platform systems** to facilitate seamless device-to-cloud communication, supporting real-time networking and distributed coordination across the platform.
- Enhanced **observability and debugging tooling** for production systems, integrating Splunk and CloudWatch metrics to reduce incident response times by 30% for critical platform issues.
- Automated **device update orchestration and rollout processes** using Python scripting and Jenkins pipelines, reducing manual deployment effort by 80% and improving deployment consistency.

SKILLS

Backend Development – Java Spring Boot, Python, NestJS, Node.js, REST APIs, Microservices.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Linux Systems.

Data & Databases – PostgreSQL, SQL, Data Warehousing, ETL, Power Query.

Systems & Reliability – Distributed Systems, Real-time Networking, Fleet Management, Observability, Incident Response.

Security & Authentication – Device Identity, Authentication Protocols, Certificate Management.

CI/CD & Tools – Jenkins, Git, Jira, Confluence.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Real-Time Device Fleet Orchestration Platform

- Architected and built a scalable backend platform for managing a fleet of 10,000+ Linux-based IoT devices, enabling real-time communication and control.
- Developed **event-driven microservices** using **Python** and **FastAPI**, processing millions of device telemetry events daily and driving automated responses for reliability and coordination.

- Implemented a **secure device authentication** mechanism leveraging mutual TLS and short-lived tokens, integrated with a **PostgreSQL** database for identity management, achieving zero unauthorized access incidents.
- Engineered a **device update orchestration system** using **Jenkins** and **Docker**, facilitating phased rollouts and automated rollback strategies that reduced deployment failures by 40%.

Financial Services Transaction Monitoring Dashboard

- Supported production operations for a critical payments operations dashboard, ensuring high availability and performance for transaction-flow monitoring.
- Developed **Python scripts** for data validation and automated alerting on failed or stalled payment messages, reducing manual oversight by 60%.
- Managed **SQL queries** and schema updates across PostgreSQL and Oracle databases to support reporting and reconciliation workflows, maintaining data integrity for audit trails.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)